CA4

November 10, 2023

[1]:
```
# Import all the tools we need
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from warnings import filterwarnings
filterwarnings(action='ignore')
```

[4]:
```
df = pd.read_csv("heart-disease.csv")
df.shape
```

[4]: (303, 14)

[5]:
```
print("Head Values:")
df.head()
```

```
Head Values:
```

[5]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	\
0	63	1	3	145	233	1	0	150	0	2.3	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	

	ca	thal	target
0	0	1	1
1	0	2	1
2	0	2	1
3	0	2	1
4	0	2	1

[6]:
```
print("Tail Values:")
df.tail()
```

```
Tail Values:
```

[6]:
```
     age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  \
298   57    0   0       140   241    0        1      123      1      0.2
299   45    1   3       110   264    0        1      132      0      1.2
300   68    1   0       144   193    1        1      141      0      3.4
301   57    1   0       130   131    0        1      115      1      1.2
302   57    0   1       130   236    0        0      174      0      0.0

     slope  ca  thal  target
298      1   0     3       0
299      1   0     3       0
300      1   2     3       0
301      1   1     3       0
302      1   1     2       0
```

[7]:
```
df.isna().sum()
```

[7]:
```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

[8]:
```
df.describe()
```

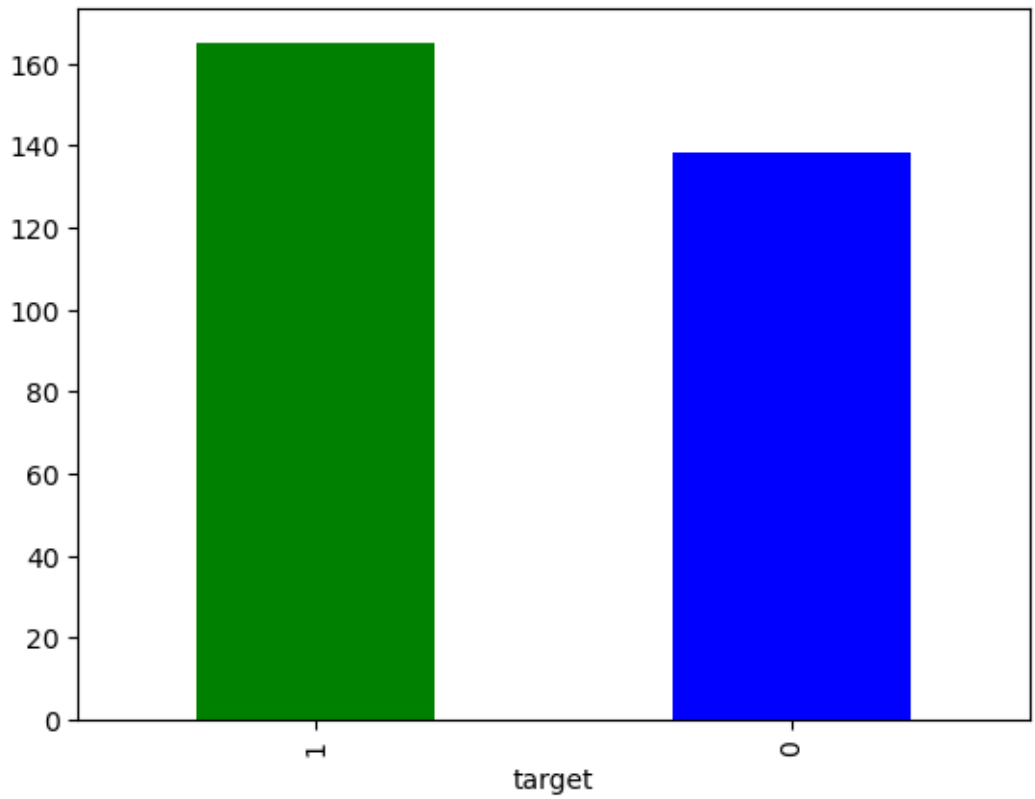
[8]:
```
              age         sex          cp    trestbps        chol         fbs  \
count  303.000000  303.000000  303.000000  303.000000  303.000000  303.000000
mean    54.366337    0.683168    0.966997  131.623762  246.264026    0.148515
std      9.082101    0.466011    1.032052   17.538143   51.830751    0.356198
min     29.000000    0.000000    0.000000   94.000000  126.000000    0.000000
25%     47.500000    0.000000    0.000000  120.000000  211.000000    0.000000
50%     55.000000    1.000000    1.000000  130.000000  240.000000    0.000000
75%     61.000000    1.000000    2.000000  140.000000  274.500000    0.000000
max     77.000000    1.000000    3.000000  200.000000  564.000000    1.000000

          restecg     thalach       exang     oldpeak       slope          ca  \
count  303.000000  303.000000  303.000000  303.000000  303.000000  303.000000
mean     0.528053  149.646865    0.326733    1.039604    1.399340    0.729373
```

```
std      0.525860   22.905161    0.469794    1.161075    0.616226    1.022606
min      0.000000   71.000000    0.000000    0.000000    0.000000    0.000000
25%      0.000000  133.500000    0.000000    0.000000    1.000000    0.000000
50%      1.000000  153.000000    0.000000    0.800000    1.000000    0.000000
75%      1.000000  166.000000    1.000000    1.600000    2.000000    1.000000
max      2.000000  202.000000    1.000000    6.200000    2.000000    4.000000

             thal      target
count  303.000000  303.000000
mean     2.313531    0.544554
std      0.612277    0.498835
min      0.000000    0.000000
25%      2.000000    0.000000
50%      2.000000    1.000000
75%      3.000000    1.000000
max      3.000000    1.000000
```

[9]:
```
# Let's find out how many of each class there
df["target"].value_counts()
```

[9]:
```
target
1    165
0    138
Name: count, dtype: int64
```

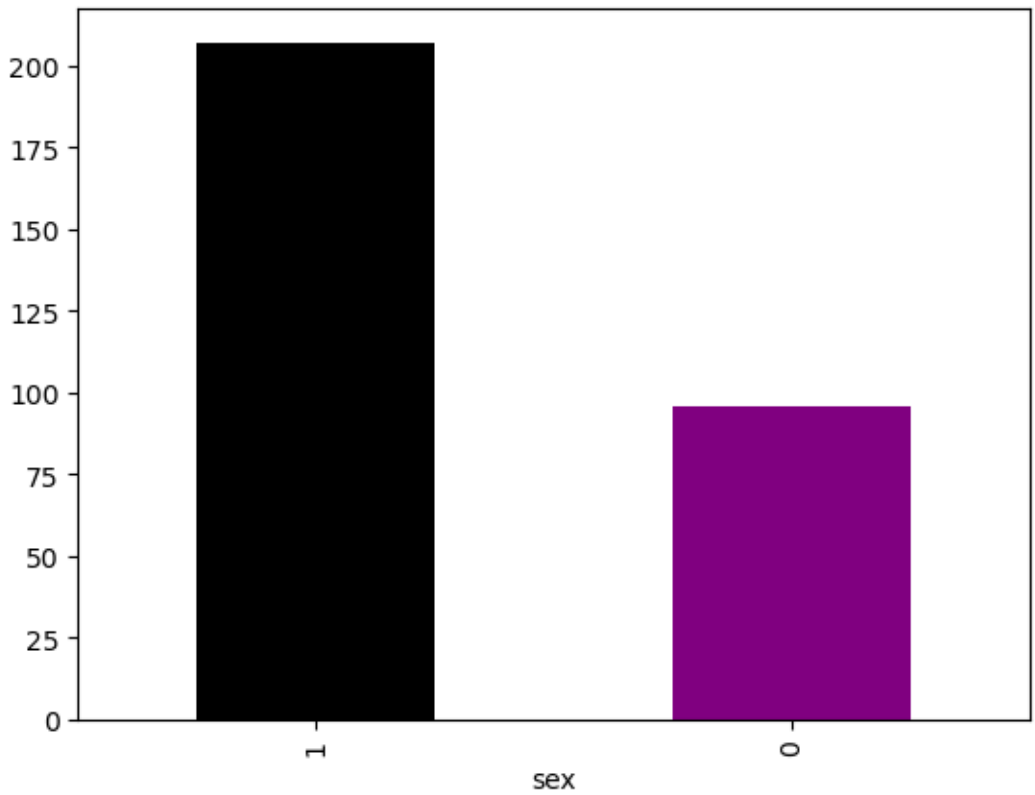
[11]:
```
df["target"].value_counts().plot(kind="bar", color=["green", "blue"]);
```



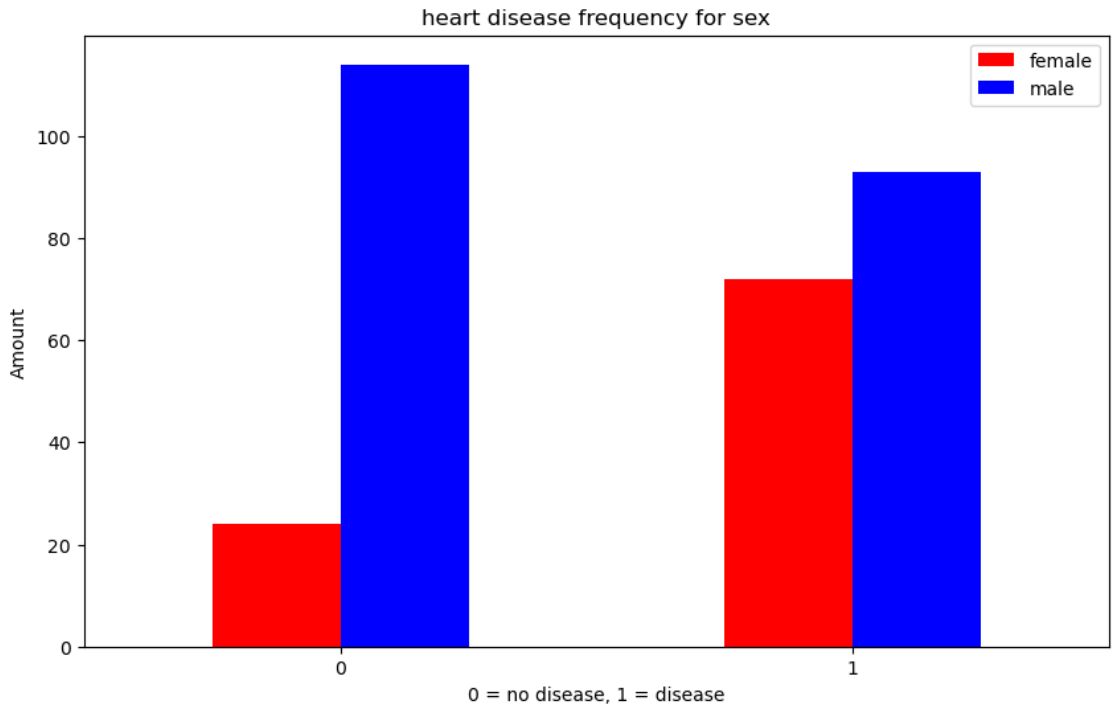
[9]:
```
df.sex.value_counts()
```

[9]:
```
1    207
0     96
Name: sex, dtype: int64
```

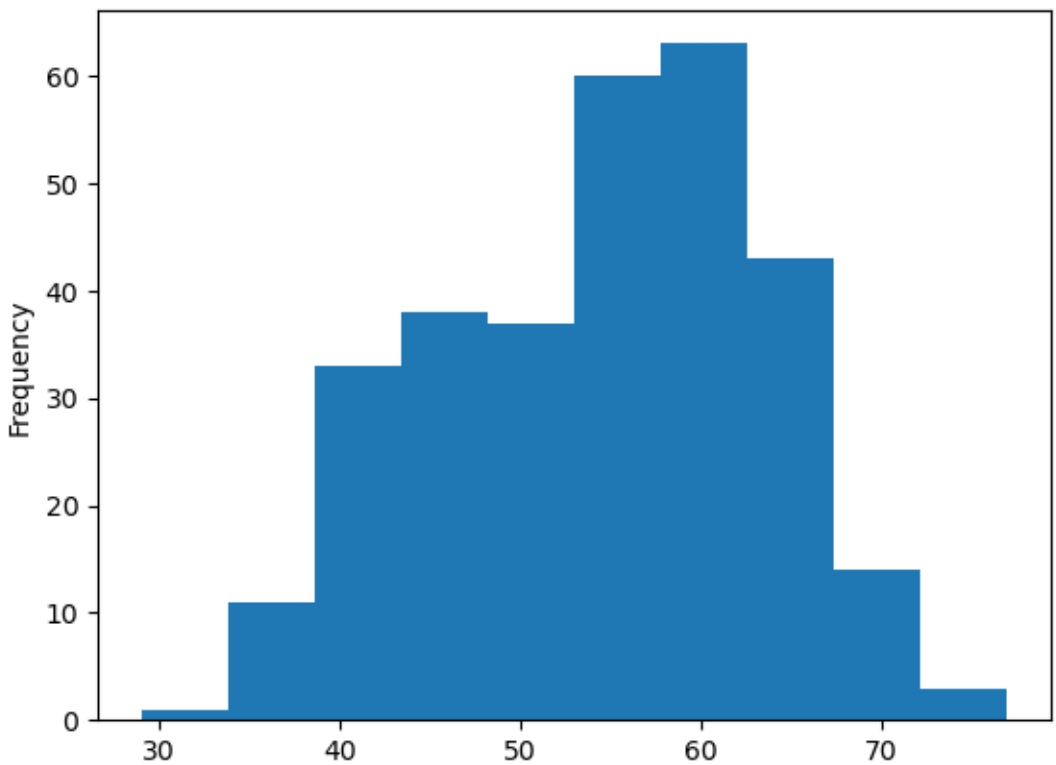
[12]:
```
df["sex"].value_counts().plot(kind="bar", color=["Black", "Purple"]);
```



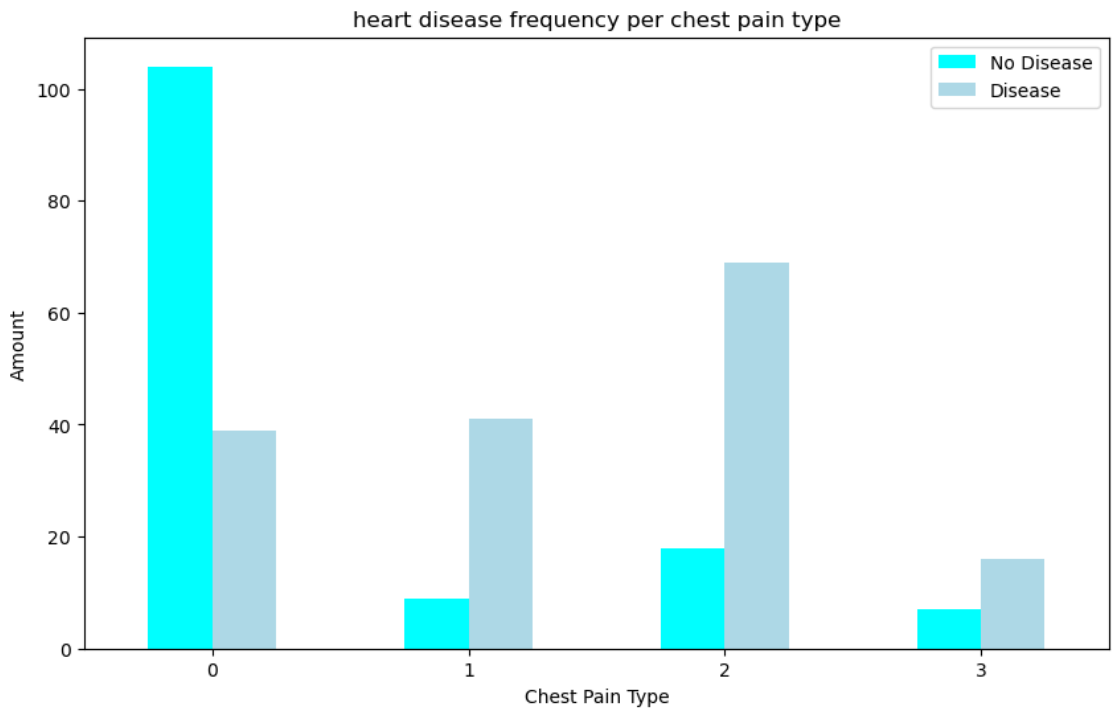
[13]:
```
pd.crosstab(df.target, df.sex).plot(kind="bar",
                                    figsize=(10,6),
                                    color=["red","blue"])
plt.title("heart disease frequency for sex")
plt.xlabel("0 = no disease, 1 = disease")
plt.ylabel("Amount")
plt.legend(["female", "male"])
plt.xticks(rotation=0);
```



[14]:
```
df.age.plot.hist();
```



[16]:
```
pd.crosstab(df.cp, df.target).plot(kind="bar",
                                   figsize=(10,6),
                                   color=["cyan","lightblue"])

plt.title("heart disease frequency per chest pain type")
plt.xlabel("Chest Pain Type")
plt.ylabel("Amount")
plt.legend(["No Disease", "Disease"])
plt.xticks(rotation=0);
```



[17]:
```
df.corr()
```

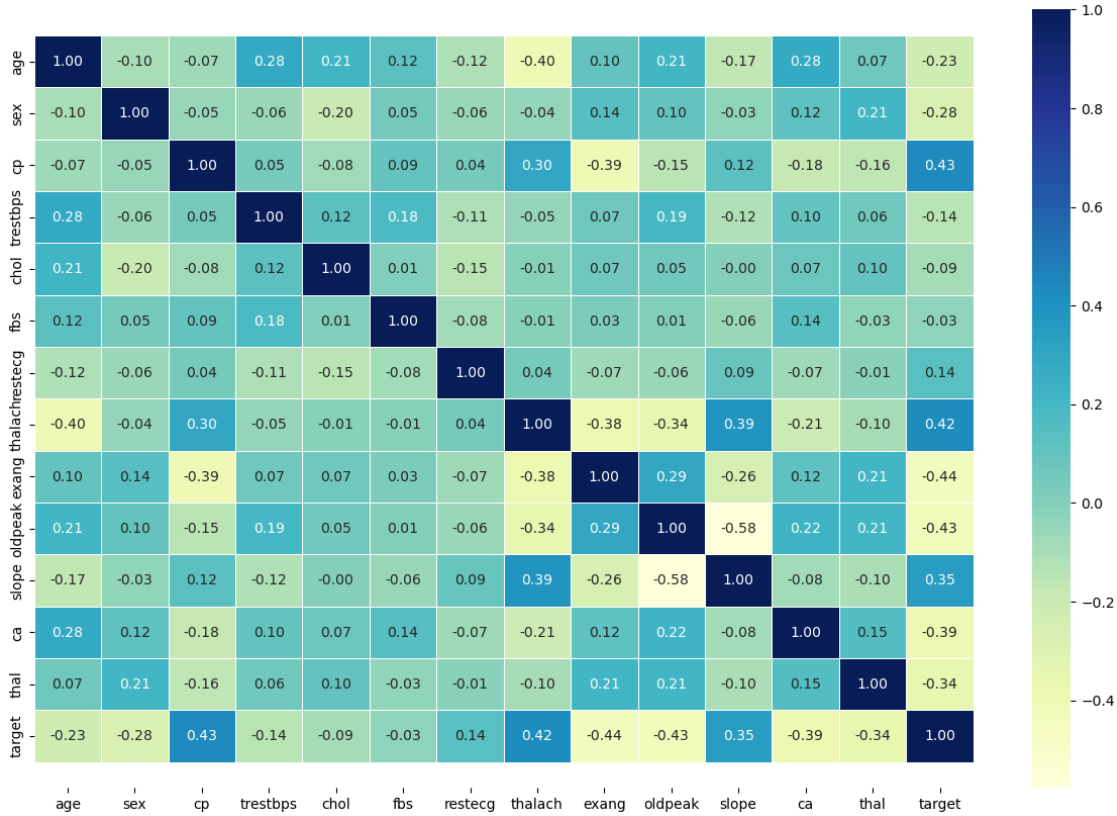
[17]:
```
               age       sex        cp  trestbps      chol       fbs  \
age       1.000000 -0.098447 -0.068653  0.279351  0.213678  0.121308
sex      -0.098447  1.000000 -0.049353 -0.056769 -0.197912  0.045032
cp       -0.068653 -0.049353  1.000000  0.047608 -0.076904  0.094444
trestbps  0.279351 -0.056769  0.047608  1.000000  0.123174  0.177531
chol      0.213678 -0.197912 -0.076904  0.123174  1.000000  0.013294
fbs       0.121308  0.045032  0.094444  0.177531  0.013294  1.000000
restecg  -0.116211 -0.058196  0.044421 -0.114103 -0.151040 -0.084189
thalach  -0.398522 -0.044020  0.295762 -0.046698 -0.009940 -0.008567
```

```
exang     0.096801  0.141664 -0.394280  0.067616  0.067023  0.025665
oldpeak   0.210013  0.096093 -0.149230  0.193216  0.053952  0.005747
slope    -0.168814 -0.030711  0.119717 -0.121475 -0.004038 -0.059894
ca        0.276326  0.118261 -0.181053  0.101389  0.070511  0.137979
thal      0.068001  0.210041 -0.161736  0.062210  0.098803 -0.032019
target   -0.225439 -0.280937  0.433798 -0.144931 -0.085239 -0.028046

           restecg   thalach     exang   oldpeak     slope        ca  \
age      -0.116211 -0.398522  0.096801  0.210013 -0.168814  0.276326
sex      -0.058196 -0.044020  0.141664  0.096093 -0.030711  0.118261
cp        0.044421  0.295762 -0.394280 -0.149230  0.119717 -0.181053
trestbps -0.114103 -0.046698  0.067616  0.193216 -0.121475  0.101389
chol     -0.151040 -0.009940  0.067023  0.053952 -0.004038  0.070511
fbs      -0.084189 -0.008567  0.025665  0.005747 -0.059894  0.137979
restecg   1.000000  0.044123 -0.070733 -0.058770  0.093045 -0.072042
thalach   0.044123  1.000000 -0.378812 -0.344187  0.386784 -0.213177
exang    -0.070733 -0.378812  1.000000  0.288223 -0.257748  0.115739
oldpeak  -0.058770 -0.344187  0.288223  1.000000 -0.577537  0.222682
slope     0.093045  0.386784 -0.257748 -0.577537  1.000000 -0.080155
ca       -0.072042 -0.213177  0.115739  0.222682 -0.080155  1.000000
thal     -0.011981 -0.096439  0.206754  0.210244 -0.104764  0.151832
target    0.137230  0.421741 -0.436757 -0.430696  0.345877 -0.391724

              thal    target
age       0.068001 -0.225439
sex       0.210041 -0.280937
cp       -0.161736  0.433798
trestbps  0.062210 -0.144931
chol      0.098803 -0.085239
fbs      -0.032019 -0.028046
restecg  -0.011981  0.137230
thalach  -0.096439  0.421741
exang     0.206754 -0.436757
oldpeak   0.210244 -0.430696
slope    -0.104764  0.345877
ca        0.151832 -0.391724
thal      1.000000 -0.344029
target   -0.344029  1.000000
```

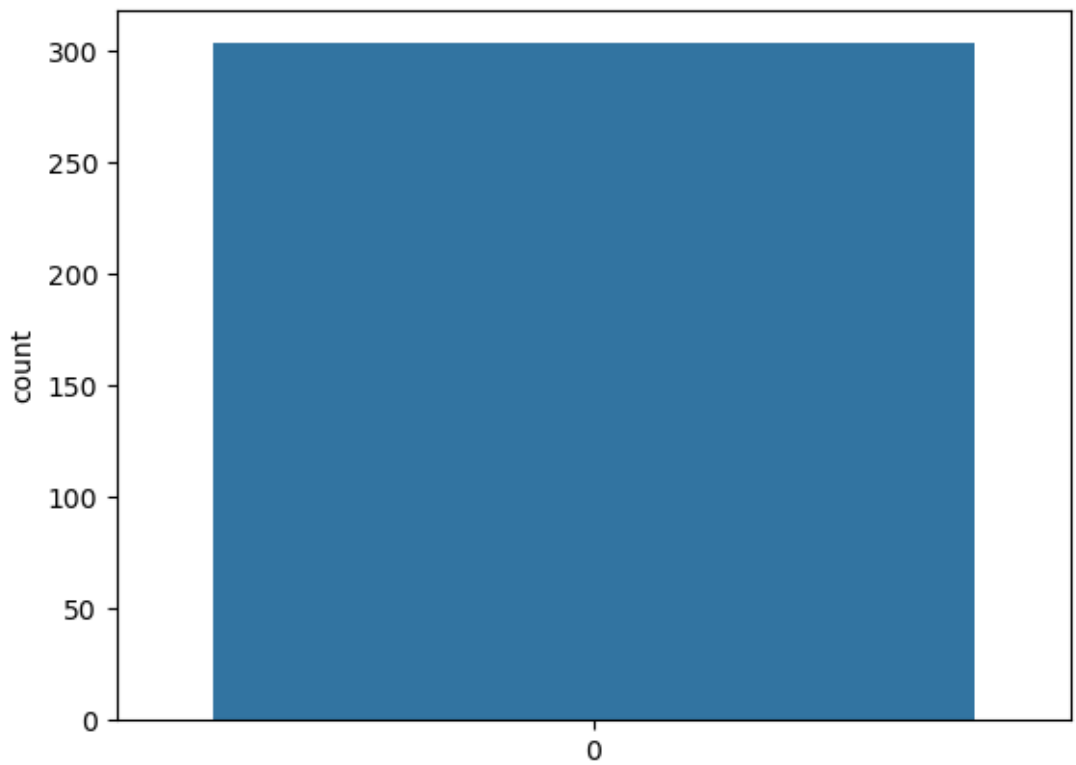
[18]:
```
corr_matrix = df.corr()
fig, ax =plt.subplots(figsize=(15,10))
ax = sns.heatmap(corr_matrix,
                 annot =True,
                 linewidths=0.5,
                 fmt=".2f",
                 cmap="YlGnBu");
bottom, top = ax.get_ylim()
```

```
ax.set_ylim(bottom+0.5,top-0.5)
```

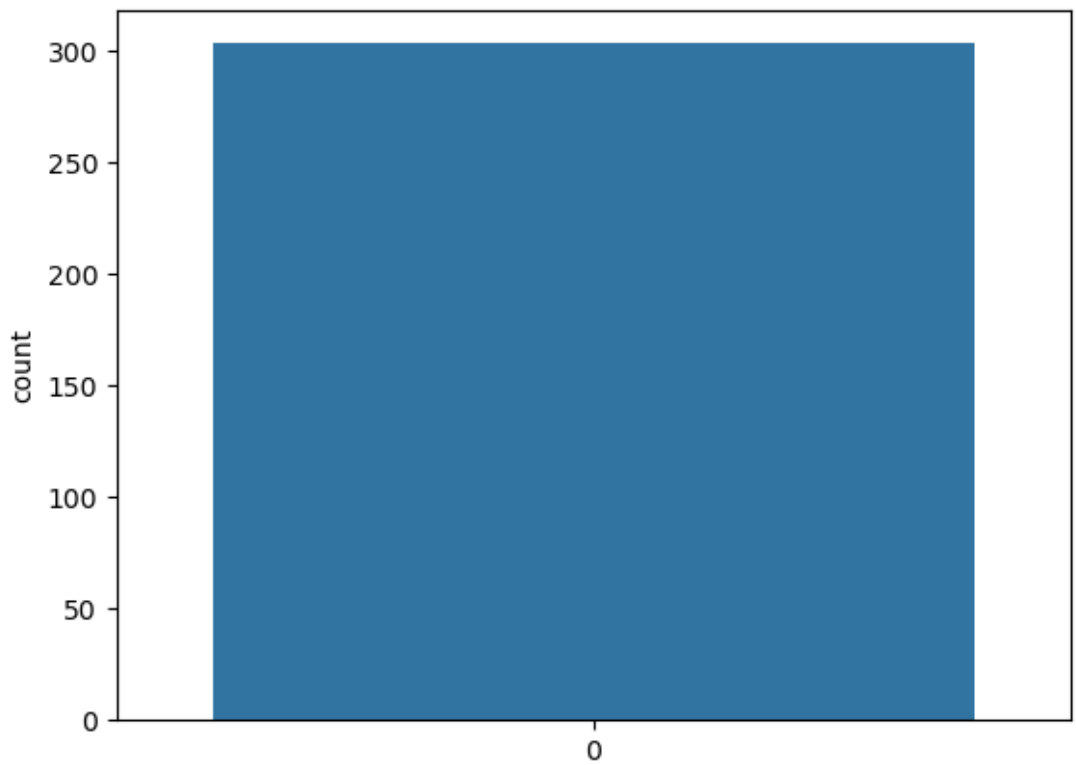
[18]: (14.5, -0.5)



[19]:
```
sns.countplot(df['age'])
plt.show()
```

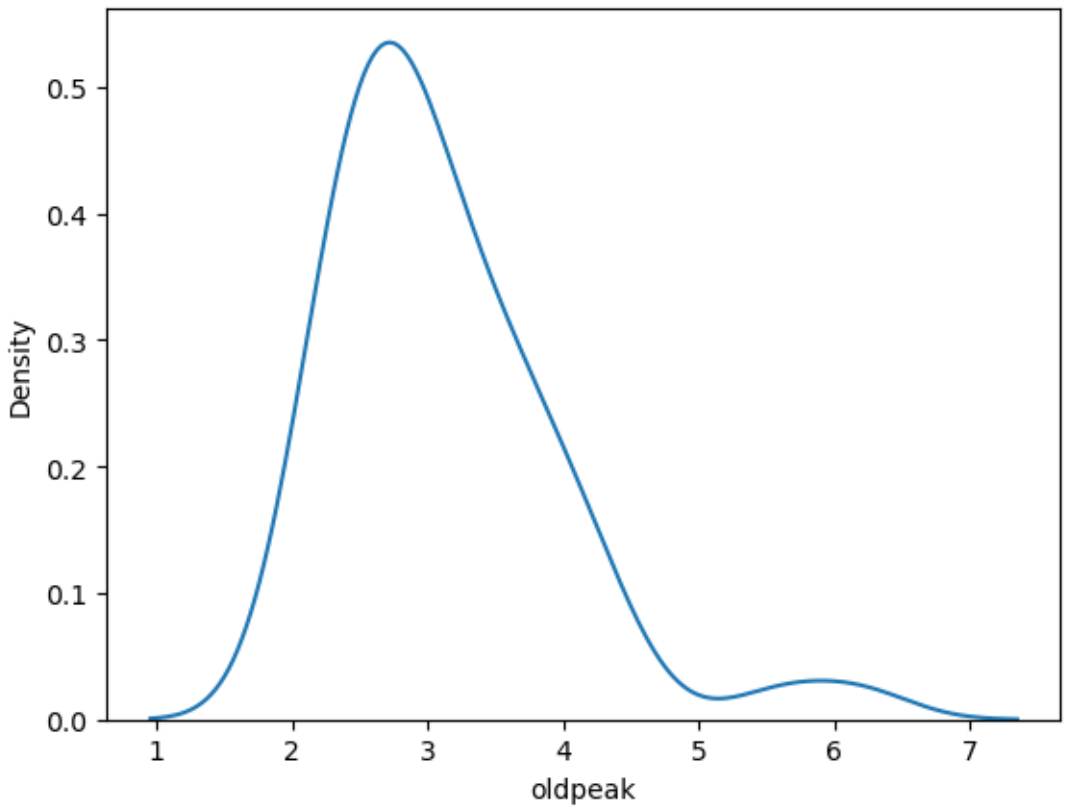


[20]:
```
sns.countplot(df['cp'])
plt.show()
```



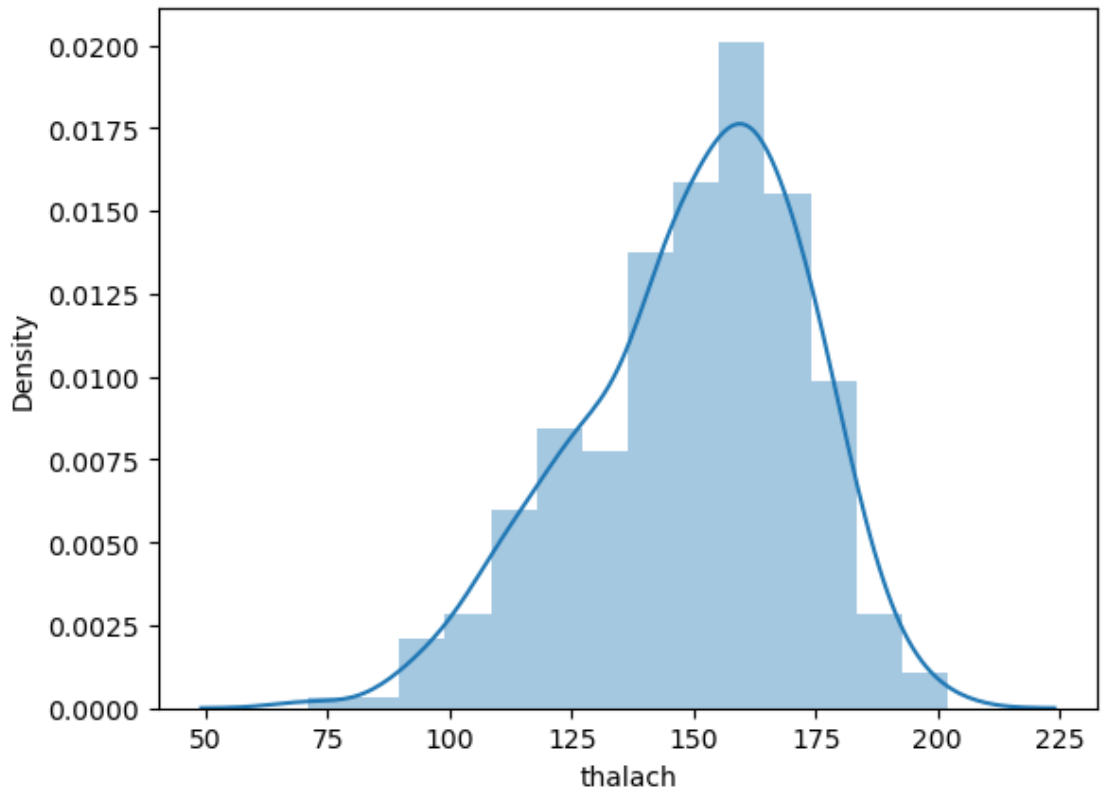
[21]:
```
sns.kdeplot(df.query('oldpeak > 2').oldpeak)
```

[21]: <Axes: xlabel='oldpeak', ylabel='Density'>



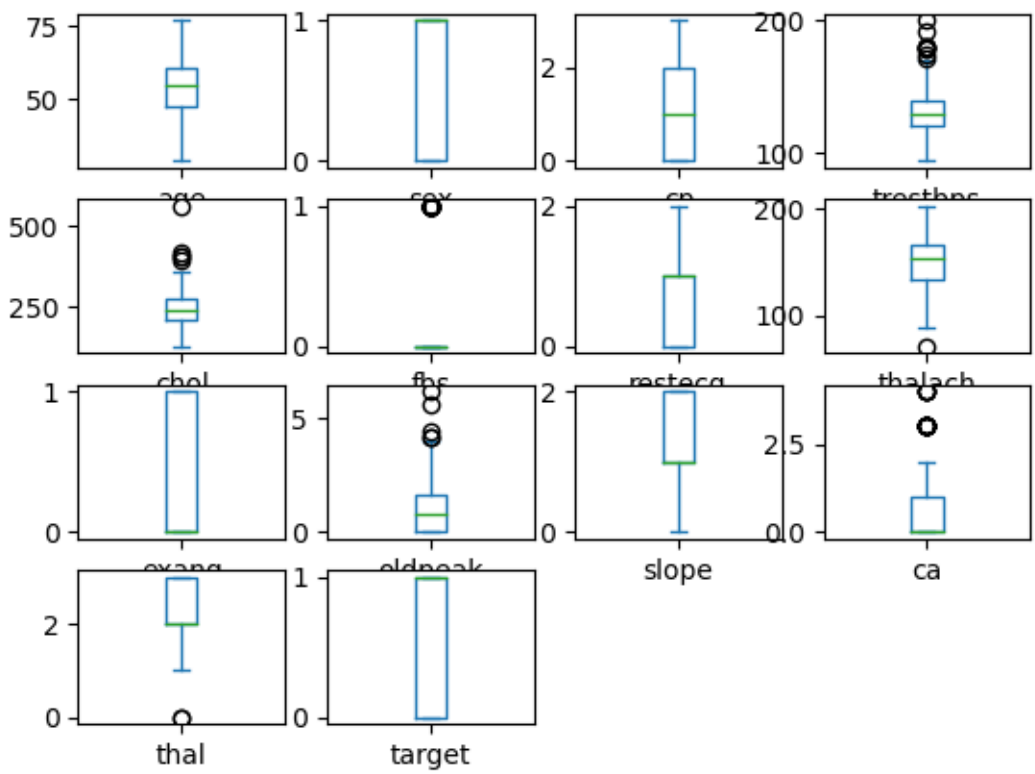
[22]:
```
sns.distplot(df['thalach'])
```

[22]: <Axes: xlabel='thalach', ylabel='Density'>



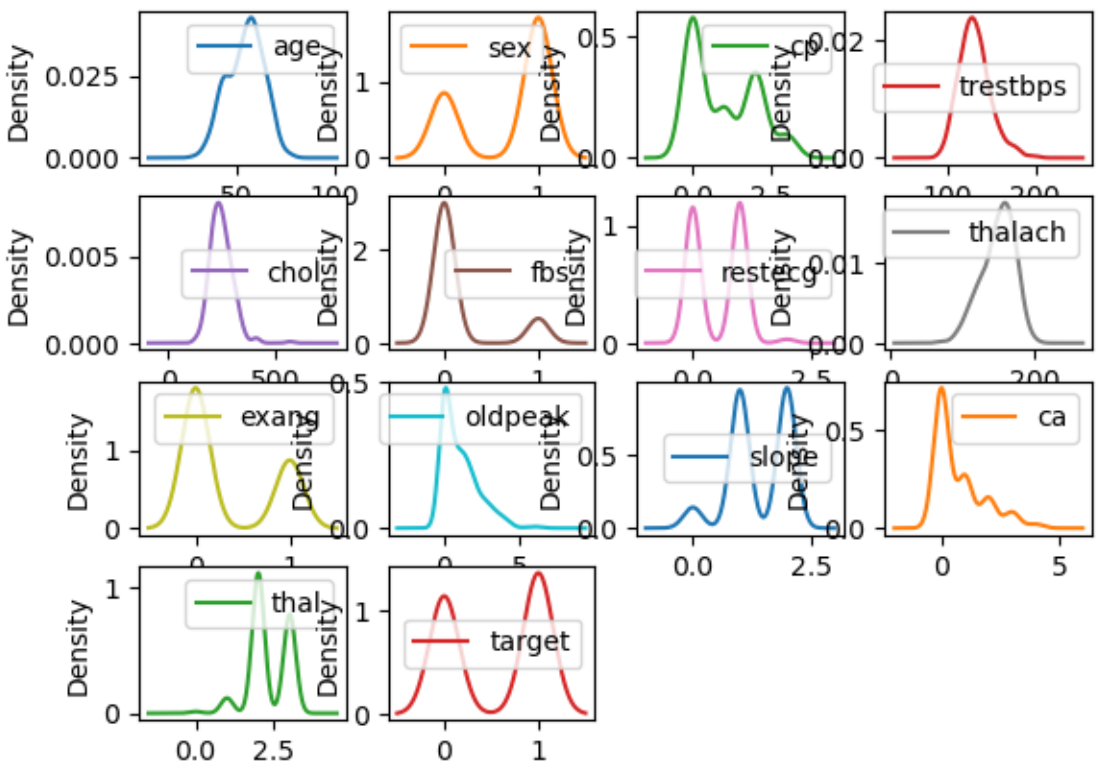
[23]:
```
df.plot(kind ='box',subplots = True, layout =(4,4),sharex = False)
```

[23]:
```
age            Axes(0.125,0.712609;0.168478x0.167391)
sex          Axes(0.327174,0.712609;0.168478x0.167391)
cp           Axes(0.529348,0.712609;0.168478x0.167391)
trestbps     Axes(0.731522,0.712609;0.168478x0.167391)
chol            Axes(0.125,0.511739;0.168478x0.167391)
fbs          Axes(0.327174,0.511739;0.168478x0.167391)
restecg      Axes(0.529348,0.511739;0.168478x0.167391)
thalach      Axes(0.731522,0.511739;0.168478x0.167391)
exang            Axes(0.125,0.31087;0.168478x0.167391)
oldpeak       Axes(0.327174,0.31087;0.168478x0.167391)
slope         Axes(0.529348,0.31087;0.168478x0.167391)
ca            Axes(0.731522,0.31087;0.168478x0.167391)
thal                Axes(0.125,0.11;0.168478x0.167391)
target           Axes(0.327174,0.11;0.168478x0.167391)
dtype: object
```

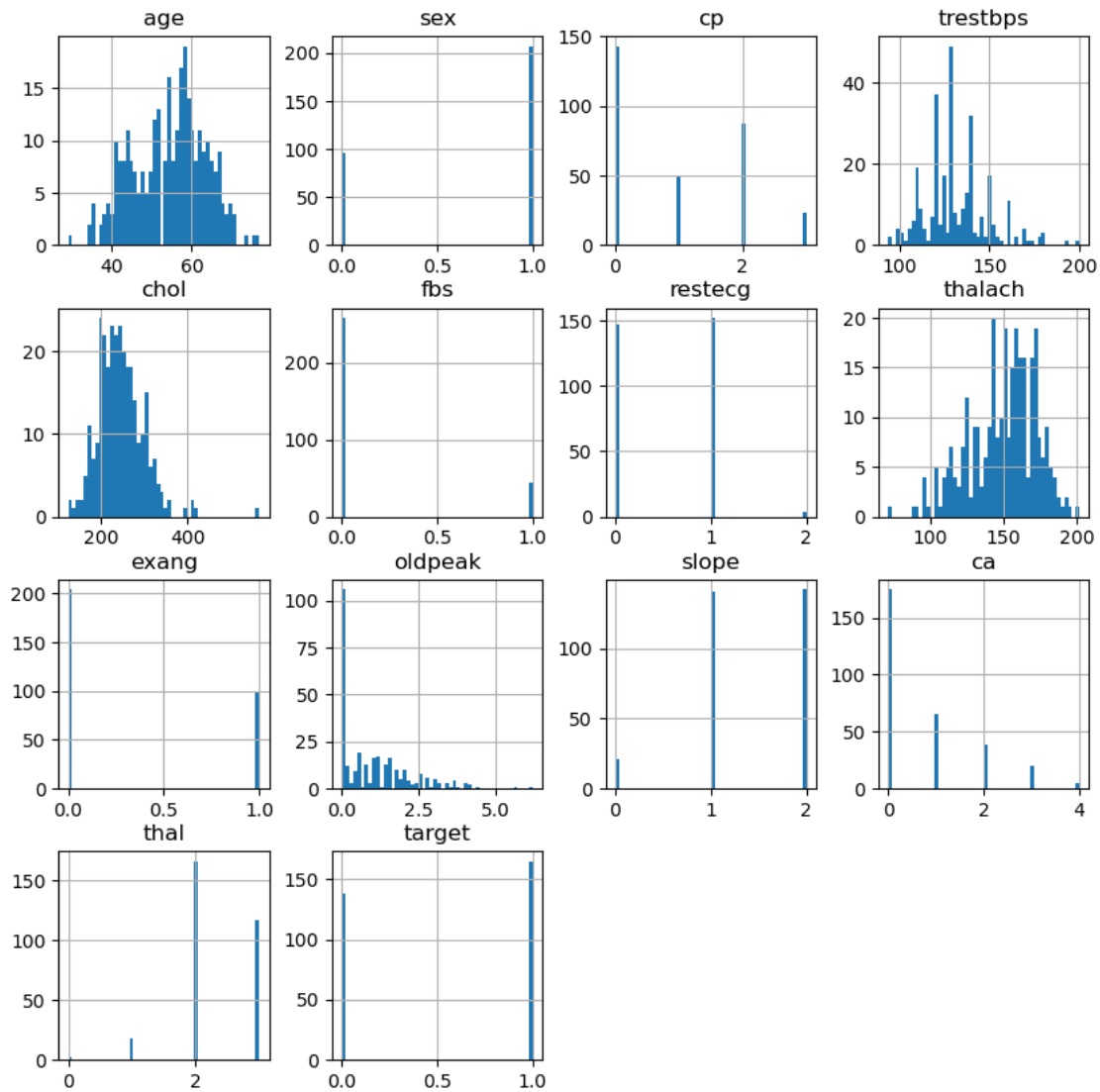


[24]:
```
df.plot(kind ='density',subplots = True, layout =(4,4),sharex = False)
```

[24]:
```
array([[<Axes: ylabel='Density'>, <Axes: ylabel='Density'>,
        <Axes: ylabel='Density'>, <Axes: ylabel='Density'>],
       [<Axes: ylabel='Density'>, <Axes: ylabel='Density'>,
        <Axes: ylabel='Density'>, <Axes: ylabel='Density'>],
       [<Axes: ylabel='Density'>, <Axes: ylabel='Density'>,
        <Axes: ylabel='Density'>, <Axes: ylabel='Density'>],
       [<Axes: ylabel='Density'>, <Axes: ylabel='Density'>,
        <Axes: ylabel='Density'>, <Axes: ylabel='Density'>]], dtype=object)
```



[25]:
```
df.hist(figsize=(10,10),bins=50)
plt.show()
```




```
[ ]: sns.pairplot(df)
```

```
[ ]: sns.violinplot(x='ca', y='age', data=df)
```

```
[25]: # split data into x and y
      X = df.drop("target",axis=1)
      Y = df["target"]
      X
```

```
[25]:      age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  \
      0     63    1   3       145   233    1        0      150      0      2.3
      1     37    1   2       130   250    0        1      187      0      3.5
      2     41    0   1       130   204    0        0      172      0      1.4
```

```
3     56    1   1       120   236    0        1       178       0       0.8
4     57    0   0       120   354    0        1       163       1       0.6
..    …   …   ..        …    …   …      …       …     …       …
298   57    0   0       140   241    0        1       123       1       0.2
299   45    1   3       110   264    0        1       132       0       1.2
300   68    1   0       144   193    1        1       141       0       3.4
301   57    1   0       130   131    0        1       115       1       1.2
302   57    0   1       130   236    0        0       174       0       0.0

     slope  ca  thal
0        0   0     1
1        0   0     2
2        2   0     2
3        2   0     2
4        2   0     2
..     …  ..   …
298      1   0     3
299      1   0     3
300      1   2     3
301      1   1     3
302      1   1     2

[303 rows x 13 columns]
```

[ ]:
```
from sklearn.model_selection import train_test_split, cross_val_score
np.random.seed(42)

X_train, X_test, Y_train, Y_test =train_test_split(X,Y,test_size=0.2)
```

[ ]:
```
X_train.head()
```

[ ]:
```
Y_train.head()
```

[ ]:
```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X_train,Y_train)
Y_pred = model.predict(X_test)

from sklearn.metrics import accuracy_score,confusion_matrix
print("Accuracy Score:",accuracy_score(Y_test,Y_pred))
```

[31]:
```
confusion_mat = confusion_matrix(Y_test,Y_pred)
print(confusion_mat)
```

```
[[25  4]
 [ 4 28]]
```

0.1 Using KNN:

[32]:
```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train,Y_train)
y_pred = model.predict(X_test)

from sklearn.metrics import accuracy_score
print("Accuracy Score:",accuracy_score(Y_test,y_pred))
```

```
Accuracy Score: 0.6557377049180327
```

0.2 Using SVC:

[33]:
```
from sklearn.svm import SVC
model = SVC()
model.fit(X_train,Y_train)
pred_y = model.predict(X_test)

from sklearn.metrics import accuracy_score
print("Accuracy Score:",accuracy_score(Y_test,pred_y))
```

```
Accuracy Score: 0.5409836065573771
```

0.3 Using DecisionTree:

[34]:
```
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(criterion='entropy',random_state=7)
model.fit(X_train,Y_train)
y_pred = model.predict(X_test)

from sklearn.metrics import accuracy_score
print("Accuracy Score:",accuracy_score(Y_test,y_pred))
```

```
Accuracy Score: 0.819672131147541
```

0.4 Using GaussianNB:

[35]:
```
from sklearn.naive_bayes import GaussianNB
model3 = GaussianNB()
model3.fit(X_train,Y_train)
y_pred3 = model3.predict(X_test)

from sklearn.metrics import accuracy_score
print("Accuracy Score:",accuracy_score(Y_test,y_pred3))
```

```
Accuracy Score: 0.8688524590163934
```

0.5 Using RandomForest:

[36]:
```
from sklearn.ensemble import RandomForestClassifier
model2 = RandomForestClassifier(random_state=1)
model2.fit(X_train, Y_train)
y_pred2 = model2.predict(X_test)

from sklearn.metrics import accuracy_score
print("Accuracy Score:",accuracy_score(Y_test,y_pred2))
```

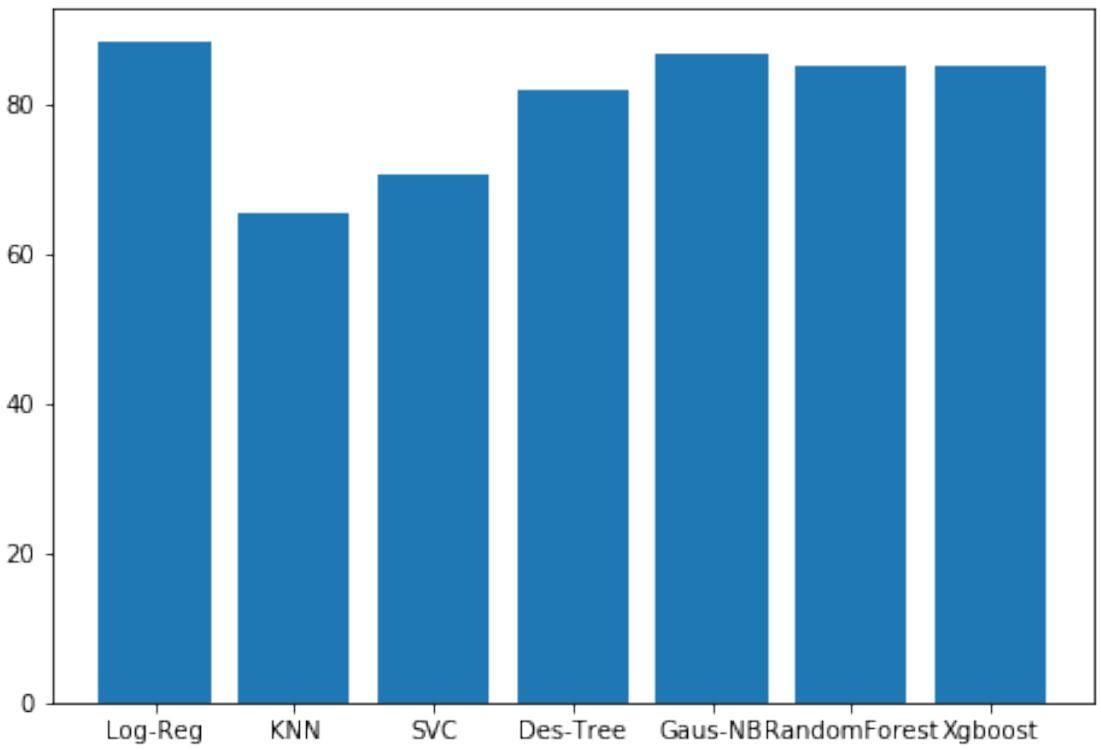
```
Accuracy Score: 0.7540983606557377
```

0.6 Using Xgboost:

[40]:
```
import xgboost as xgb
model5 = xgb.XGBClassifier(random_state=1)
model5.fit(X_train, Y_train)
y_pred5 = model5.predict(X_test)

from sklearn.metrics import accuracy_score
print("Accuracy Score:",accuracy_score(Y_test,y_pred5))
```

```
[15:53:47] WARNING: C:/Users/Administrator/workspace/xgboost-
win64_release_1.3.0/src/learner.cc:1061: Starting in XGBoost 1.3.0, the default
evaluation metric used with the objective 'binary:logistic' was changed from
'error' to 'logloss'. Explicitly set eval_metric if you'd like to restore the
old behavior.
Accuracy Score: 0.819672131147541
```

[41]:
```
fig = plt.figure()
ax = fig.add_axes([0,0,1,1])
langs = ['Log-Reg', 'KNN', 'SVC', 'Des-Tree', 'Gaus-NB', 'RandomForest',␣
 ↪'Xgboost']
students = [88.52,65.57,70.49,81.96,86.88,85.24,85.24]
ax.bar(langs,students)
plt.show()
```



1 Hence we will use LogisticRegression algorithms for training my model.